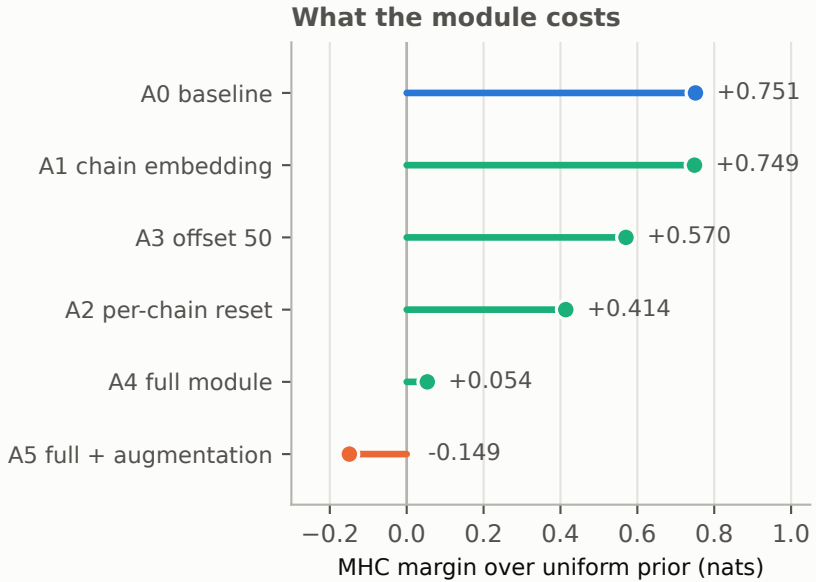
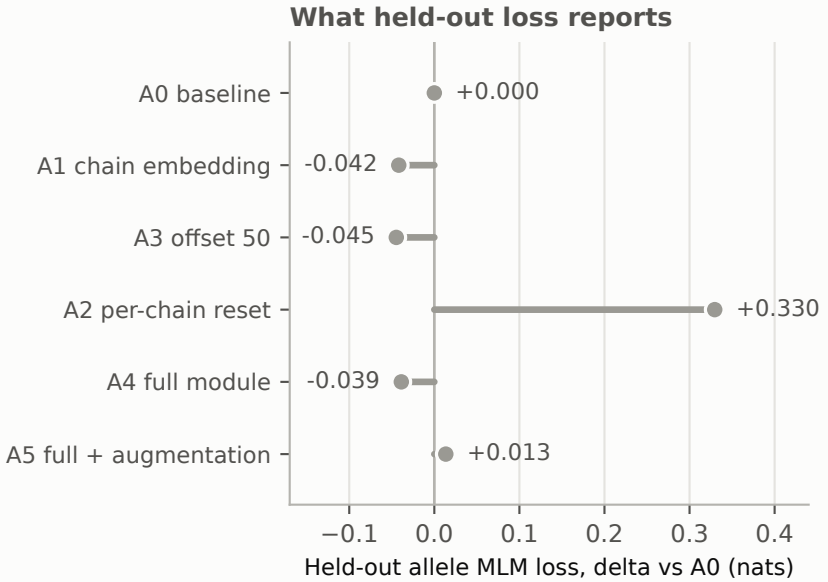
Held-out loss calls the full module free while it spends the scorer's whole margin

ESM-2 650M, LoRA r=16, 12,172,800 trainable parameters in every arm, 300 steps, one seed. Margin is the MHC score above a uniform prior over 20 amino acids; below zero the scorer is worse than guessing.





A4 is 0.039 nats better than baseline on held-out-allele loss and keeps 0.054 of the baseline's 0.751 margin. The cost tracks how much of the module is enabled: the chain embedding is free, per-chain reset is the most expensive single component.
One seed per arm, so the ordering is suggestive and the gaps carry no interval.